

Subject Description Form

Subject Code	COMP5111
Subject Title	Database Systems and Management
Credit Value	3
Level	5
Pre-requisite/ Exclusion	Nil
Objectives	The objectives of this subject are to enable students to: 1. gain a good understanding of the architecture and functioning of database management systems, as well as the associated tools and techniques; 2. understand and be able to apply the principles and practices of good database design; 3. appreciate the direction of database technology and their implication on management and planning of database systems; 4. appraise and use alternative conceptual and/or data models for documenting enterprise databases; 5. evaluate available DBMS systems against organisation needs and negotiate the acquisition of DBMS.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) design database solutions to solve common management/business problems, leveraging emerging technologies for global challenges; b) evaluate the effectiveness of specific database solutions in solving management problems; and c) articulate the organisational impact of database solutions, demonstrating critical awareness and integrity in making responsible decisions.
Subject Synopsis/ Indicative Syllabus	• Overview of Database Management and Architecture • Relational DBMS: Entity-relationship (ER) modelling, Relational database design, SQL and relational algebra, View mechanisms. • DB Implementation and Operational Issues: Data dependencies and normalisation, Query processing and optimisation, Security and integrity constraints, Physical database design, Transactions, recovery and concurrency issues, Commercial DBMSs. • Selected Topics for Database Management: Database administration, Database applications for enterprises, Database project development. • Selected Topics for Database Technology: Object-oriented and semantic data modelling, Distributed database architecture, high-dimensional databases, Web databases.
Teaching/Learning Methodology	Class activities including - lecture, tutorial, lab, workshop seminar where applicable.
Assessment Methods in Alignment with Intended Learning Outcomes	

	Specific Assessment Methods/Tasks	% weighting	Intended subject learning outcomes to be assessed		
			a	b	c
	Assignments, Tests & Projects	30	✓	✓	✓
	Final Examination	70	✓	✓	✓
	Total	100			
Student study effort expected	Class Contact:				
	Class activities (lecture, tutorial, lab)			39 hours	
	Other student study effort:				
	Assignments, Quizzes, Projects, Exams			66 hours	
	Total student study effort			105 hours	
Reading list and references	(1) Michael V. Mannino. Database Design, Application Development, & Administration, 5th edition, McGraw-Hill, 2011.				
	(2) David M. Kroenke. Database Processing: Fundamentals, Design and Implementation, 13/E, Prentice Hall, 2013.				
	(3) Avi Silberschatz, Henry F. Korth, S. Sudarshan. Database System Concepts 6th Edition. McGraw Hill, 2010.				
	(4) Mamoulis, Nikos. Spatial data management. Springer Nature, 2022.				